Improved lattice-packing bounds in small dimensions

Nina Zubrilina

October 26, 2026

Abstract

We strengthen the upper bound on the density of a lattice packing in small dimensions. A lattice cannot have too many short vectors; we modify the Cohn–Elkies linear program accordingly by allowing the auxiliary function to be positive on a short interval beyond the minimum distance. For $11 \le n \le 16$, our bound is tighter than the three-point bound of Cohn, de Laat and Salmon. For $n \in [17, 23] \cup [25, 40] \cup [43, 57] \cup [62, 75] \cup [83, 91]$, our bound lies below the published numerical values of the Cohn–Elkies bound.

1 Introduction

Let $\Lambda \subset \mathbb{R}^n$ be a full-rank lattice normalized to have shortest nonzero vector of length 1, and write $\rho(\Lambda) = 1/\operatorname{vol}(\mathbb{R}^n/\Lambda)$. Placing balls of radius $1/2$ at the lattice points produces a packing of density $v_n 2^{-n}\rho(\Lambda)$, where $v_n = \pi^{n/2}/\Gamma(n/2+1)$ is the unit ball volume. The optimal lattice packing density is

$$\Delta_n^{\mathrm{lat}} = \frac{v_n}{2^n} \sup_{\Lambda} \rho(\Lambda).$$

The Cohn–Elkies bound [2] controls density through an auxiliary function with nonnegative Fourier transform. In the lattice case, Poisson summation bounds $\rho(\Lambda)\widehat{f}(0)$ by $\sum_{v\in\Lambda} f(v)$, and the condition $f(x) \le 0$ for $|x| \ge 1$ makes the latter sum at most $f(0)$.

For a lattice packing, we obtain additional constraints on which short vectors can cooccur. Martinet [5] observed that each class of $\Lambda/2\Lambda$ contains at most one pair $\{\pm v\}$ of length less than $\sqrt{2}$; pairs with lengths slightly above $\sqrt{2}$ have nearly orthogonal directions, which allows us to bound their count as well. These restrictions allow us to bound the total positive contribution from a class.

Our modification always does at least as well as the Cohn–Elkies program, and in fact one can show that for n sufficiently large, the two programs are equivalent. The improvement we get is in small dimensions, with diminishing returns as n grows. Further refinements of the class-counting constraints, including relations among classes modulo higher powers of 2, give small additional improvements; we omit those in this paper as they do not substantially narrow the gap to the best known lattice densities.

Section 2 states and proves the resulting bound. Section 3 describes the polynomial certificates and the method of their verification. Finally, Section 4 presents the certified bounds and numerical comparisons.

The numerical exploration of the linear program was performed by GPT-6 Astra.

2 The modified linear program

Let $n \geq 2$ and $\sqrt{2} \leq s < \min\{2, \sqrt{2/(1-n^{-1/2})}\}$. For a nonzero class C of $\Lambda/2\Lambda$ and $t \geq 1$, we want to bound

$$N_C(t) := \frac{1}{2}\#\{v \in C : |v| < t\},$$

the number of opposite pairs in C of length less than t. We will bound $N_C(s)$ by $B(s)$, and, when C contains a pair shorter than a, bound the number of remaining pairs shorter than s by $A(a,s)$. To that end, let $c(s) = 1 - 2/s^2$, so that $0 \leq c(s) < 1/\sqrt{n}$, and define

$$B(s) = \left\lfloor \frac{n(1-c(s)^2)}{1-nc(s)^2} \right\rfloor. \tag{1}$$

For $1 \leq a \leq \sqrt{2}$, set $A(a,s) = 0$ when $a^2 + s^2 \leq 4$. Otherwise put

$$c_a = \frac{a^2+s^2-4}{2as}, \qquad c_a' = \frac{c(s)+c_a^2}{1-c_a^2},$$

and define

$$A(a,s) = \begin{cases} \min\left\{B(s)-1, \left\lfloor \dfrac{(n-1)(1-(c_a')^2)}{1-(n-1)(c_a')^2} \right\rfloor\right\}, & (n-1)(c_a')^2 < 1, \\ B(s)-1, & (n-1)(c_a')^2 \geq 1. \end{cases} \tag{2}$$

We will use the following class of step functions G. Choose an integer M and partition points $1 = e_0 < \cdots < e_M < \min\{2, \sqrt{2/(1-n^{-1/2})}\}$, with $e_J = \sqrt{2}$ for some $0 < J < M$. Given heights $g_0, \ldots, g_{M-1} \geq 0$, define the step function $G : [0,\infty) \to [0,\infty)$ by

$$G(t) = \begin{cases} g_b, & e_b \leq t < e_{b+1}, \quad 0 \leq b < M, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

We require $g_J \geq \cdots \geq g_{M-1}$ and set $g_M = 0$; no monotonicity is imposed on the heights below $\sqrt{2}$.

We define the *G-budget* $\kappa(G)$ by

$$\begin{aligned} \kappa_\emptyset &= \sum_{b=J}^{M-1} (g_b - g_{b+1}) B(e_{b+1}), \\ \kappa_j &= g_j + \sum_{b=J}^{M-1} (g_b - g_{b+1}) A(e_{j+1}, e_{b+1}) \quad (0 \leq j < J), \\ \kappa(G) &= \max\{\kappa_\emptyset, \kappa_0, \ldots, \kappa_{J-1}\}. \end{aligned} \tag{4}$$

As proved below, this is a uniform upper bound on the total G-weight of opposite pairs in any nonzero class C of $\Lambda/2\Lambda$:

$$\frac{1}{2}\sum_{v \in C} G(|v|) \leq \kappa(G).$$

The factor $1/2$ counts each opposite pair once. We normalize the Fourier transform as

$$\widehat{h}(y) = \int_{\mathbb{R}^n} h(x) e^{-2\pi i \langle x, y \rangle}\, dx.$$

Theorem 1. *Let $\Lambda \subset \mathbb{R}^n$ be a full-rank lattice of minimum length 1, and let G be as above. Suppose h is a real Schwartz function such that $\widehat{h} \geq 0$, $\widehat{h}(0) > 0$, and*

$$h(x) \leq G(|x|) \quad (1 \leq |x| < e_M), \qquad h(x) \leq 0 \quad (|x| \geq e_M).$$

Then

$$\rho(\Lambda) \leq \frac{h(0) + 2(2^n - 1)\kappa(G)}{\widehat{h}(0)}. \tag{5}$$

Proof. Following Cohn–Elkies, Poisson summation gives

$$\rho(\Lambda)\widehat{h}(0) \leq \rho(\Lambda) \sum_{w \in \Lambda^*} \widehat{h}(w) = \sum_{v \in \Lambda} h(v)$$

and our goal is to show

$$\sum_{v \in \Lambda \setminus \{0\}} h(v) \leq 2(2^n - 1)\kappa(G).$$

Suppose nonzero vectors $v, w \in \Lambda$ belong to the same class in $\Lambda/2\Lambda$, with $v \neq \pm w$. Then $(v + w)/2$ and $(v - w)/2$ are in $\Lambda \setminus \{0\}$, hence

$$|v \pm w| \geq 2, \qquad |\langle v, w \rangle| \leq \frac{|v|^2 + |w|^2 - 4}{2}. \tag{6}$$

In particular, v and w cannot both have length below $\sqrt{2}$.

The function $F(x, y) = (x^2 + y^2 - 4)/(2xy)$ increases in each variable for $1 \leq x, y < 2$. Thus, for $|v|, |w| < s$, $v \neq \pm w$,

$$|\langle v/|v|, w/|w| \rangle| \leq F(|v|, |w|) \leq F(s, s) = c(s).$$

Now, let $u_1, \ldots, u_N \in \mathbb{R}^d$ be unit column vectors with $|\langle u_i, u_j \rangle| \leq c$ for $i \neq j$ and $dc^2 < 1$. The positive semidefinite matrix $S = \sum_i u_i u_i^T$ satisfies $\operatorname{tr} S = N$ and $\operatorname{tr}(S^2) = \sum_{i,j} \langle u_i, u_j \rangle^2$. Cauchy–Schwarz applied to its d eigenvalues gives

$$N^2 = (\operatorname{tr} S)^2 \leq d \operatorname{tr}(S^2) \leq d\big(N + N(N-1)c^2\big)$$

so $N \leq d(1 - c^2)/(1 - dc^2)$. Hence

$$N_C(s) \leq B(s). \tag{7}$$

Now suppose $N_C(a) = 1$ for some $1 \leq a \leq \sqrt{2}$, and let $\{\pm v\}$ be this short pair. If $a^2 + s^2 \leq 4$, inequality (6) excludes any additional pair of length less than s. Otherwise, let $u = v/|v|$, and choose one vector v_i from each of the $N_C(s) - 1$ other opposite pairs of length less than s in C. Monotonicity of F gives $|\langle u_i, u \rangle| \leq F(a, s) = c_a < 1$ for $u_i = v_i/|v_i|$; here $c_a < 1$ follows from $(s - a)^2 < 4$. For $\eta_i = \langle u_i, u \rangle$, the normalized orthogonal projection of u_i onto $u^\perp$ is $z_i = (u_i - \eta_i u)/\sqrt{1 - \eta_i^2}$. For $i \neq \ell$,

$$|\langle z_i, z_\ell \rangle| = \frac{|\langle u_i, u_\ell \rangle - \eta_i \eta_\ell|}{\sqrt{(1 - \eta_i^2)(1 - \eta_\ell^2)}} \leq \frac{c(s) + c_a^2}{1 - c_a^2} = c_a'.$$

The projections are well defined because $|\eta_i| < 1$. Applying the relative bound for $d = n - 1$ for $u^\perp$ when $(n - 1)(c_a')^2 < 1$, and combining it with $N_C(s) - 1 \leq B(s) - 1$ from (7), gives

$$N_C(s) - 1 \leq A(a, s) \qquad \text{if } N_C(a) = 1. \tag{8}$$

When $(n-1)(c'_a)^2 \geq 1$, only the bound $B(s)-1$ is used; when $a^2+s^2 \leq 4$, the left side is zero.

Finally, we bound the sum of the auxiliary function over each nonzero class C of $\Lambda/2\Lambda$. Let R_C contain one representative of each opposite pair in C, and call a pair short if its length is less than $\sqrt{2}$. Let m_ℓ count the pairs with lengths in $[e_\ell, e_{\ell+1})$ for $J \leq \ell < M$, and put $N_b = \sum_{\ell=J}^{b} m_\ell = N_C(e_{b+1}) - N_C(\sqrt{2})$. Summation by parts gives

$$\sum_{\ell=J}^{M-1} g_\ell m_\ell = \sum_{b=J}^{M-1} (g_b - g_{b+1}) N_b.$$

Since $g_b - g_{b+1} \geq 0$, we may substitute upper bounds for N_b. If C has no short pair, then $N_C(\sqrt{2}) = 0$, and (7) gives $N_b = N_C(e_{b+1}) \leq B(e_{b+1})$, so

$$\sum_{v \in R_C} G(|v|) = \sum_{\ell=J}^{M-1} g_\ell m_\ell \leq \sum_{b=J}^{M-1} (g_b - g_{b+1}) B(e_{b+1}) = \kappa_\emptyset.$$

If its unique short pair lies in $[e_j, e_{j+1})$, then $N_C(e_{j+1}) = N_C(\sqrt{2}) = 1$, and (8) gives $N_b = N_C(e_{b+1}) - 1 \leq A(e_{j+1}, e_{b+1})$. Including that pair gives

$$\sum_{v \in R_C} G(|v|) = g_j + \sum_{\ell=J}^{M-1} g_\ell m_\ell \leq g_j + \sum_{b=J}^{M-1} (g_b - g_{b+1}) A(e_{j+1}, e_{b+1}) = \kappa_j.$$

Thus

$$\sum_{v \in C} h(v) \leq 2 \sum_{v \in R_C} G(|v|) \leq 2\kappa(G).$$

Every nonzero vector in 2Λ has length at least $2 > e_M$ and hence contributes nonpositively, so summing over the $2^n - 1$ nonzero classes and applying the opening Poisson inequality gives $\rho(\Lambda)\widehat{h}(0) \leq h(0) + 2(2^n - 1)\kappa(G)$, which proves the theorem. □

3 Exact polynomial certificates

For each dimension, we provide a *certificate*: rational data that a verification program, or *verifier*, checks using exact arithmetic to establish an upper bound on lattice packing density in $\mathbb{R}^n$ via Theorem 1.

We construct radial auxiliary functions using the generalized Laguerre polynomials $L_k^{(\alpha)}$. Set $\alpha = n/2 - 1$ and write

$$p(t) = \sum_{k=0}^{K} c_k L_k^{(\alpha)}(t), \qquad q(t) = \sum_{k=0}^{K} (-1)^k c_k L_k^{(\alpha)}(t).$$

For $f(x) = p(2\pi|x|^2)e^{-\pi|x|^2}$, the Fourier transform is $\widehat{f}(y) = q(2\pi|y|^2)e^{-\pi|y|^2}$, by the Fourier eigenfunction identity in [2, Section 7].

A certificate for a density bound consists of rational numbers c_k, $\tau_0 > 0$, $E_b = e_b^2$, and $\Gamma_b \geq 0$. Set $\tau_b = \tau_0 E_b$ and define a rescaling

$$h(x) = f\left(\sqrt{\frac{\tau_0}{2\pi}}\, x\right) = p(\tau_0|x|^2)e^{-\tau_0|x|^2/2}.$$

Fourier scaling gives

$$\widehat{h}(y) = \left(\frac{2\pi}{\tau_0}\right)^{n/2} \widehat{f}\left(\sqrt{\frac{2\pi}{\tau_0}}\, y\right).$$

The annulus $e_b \le |x| \le e_{b+1}$ now corresponds to $\tau_b \le \tau_0|x|^2 \le \tau_{b+1}$.

The verifiers check the partition-point conditions from Section 2 together with the following polynomial inequalities

$$\begin{aligned} q(t) &> 0 && (t \ge 0), \\ \Gamma_b\left(1 + \frac{t-\tau_b}{2} + \frac{(t-\tau_b)^2}{8}\right) - p(t) &\ge 0 && (\tau_b \le t \le \tau_{b+1}), \\ p(t) &< 0 && (t \ge \tau_M). \end{aligned} \tag{9}$$

The middle condition is imposed for every $0 \le b < M$, including both endpoints. With $u = (t-\tau_b)/2 \ge 0$, the inequality $1 + u + u^2/2 \le e^u$ shows that it implies

$$p(t)e^{-t/2} \le \Gamma_b e^{-\tau_b/2}.$$

Thus these polynomial inequalities imply the required pointwise bounds on h, as well as strict positivity of $\widehat{h}$.

To evaluate the budget $\kappa(G)$ in (4) using rational arithmetic, put

$$R_b = \Gamma_b \left(\sum_{j=0}^{512} \frac{(\tau_b/2)^j}{j!}\right)^{-1}, \qquad g_b = \begin{cases} R_b, & b < J, \\ \max_{b \le \ell < M} R_\ell, & b \ge J. \end{cases}$$

The truncated exponential series gives $R_b \ge \Gamma_b e^{-\tau_b/2}$, and the backward maxima ensure the required monotonicity without decreasing any height. Moreover, (1)–(2) can be evaluated exactly from the squared partition points: for example, $c_a^2 = (a^2 + s^2 - 4)^2/(4a^2s^2)$ is rational. Consequently the resulting κ and

$$\rho_+ = \frac{p(0) + 2(2^n - 1)\kappa}{q(0)}$$

are rational. Since $\widehat{h}(0) = (2\pi/\tau_0)^{n/2} q(0)$, Theorem 1 gives $\rho(\Lambda) \le (\tau_0/(2\pi))^{n/2}\rho_+$. Thus

$$\Delta_n^{\mathrm{lat}} \le \frac{\tau_0^{n/2}\rho_+}{8^{n/2}\Gamma(n/2+1)}. \tag{10}$$

The reported decimal bounds are tested exactly. For even n, the density expression in (10) is rational. For odd $n = 2m+1$, it is

$$\frac{2\tau_0^m \rho_+}{4^m (2m+1)!!}\sqrt{\frac{\tau_0}{8\pi}}.$$

Choose a rational number $0 < \xi < \pi$. A positive proposed density bound in odd dimension is checked by squaring after replacing π by ξ. Positivity of each proposed decimal is checked before squaring. The lower bound ξ is obtained from $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ using the appropriate upper and lower alternating Taylor sums.

Floating-point linear programming supplies candidate coefficients; the exact sign tests and decimal comparisons establish their validity. Following the data-and-verifier presentation of Cohn–Kumar [4], the exact certificates and verification code are available in the GitHub repository. Four independent implementations use real-root counting, integer Taylor bounds, Bernstein coefficients, and Descartes' rule of signs; the last applies to even dimensions. Each reconstructs the polynomials, counting bounds, and budget from the certificate data.

4 Computational results

Table 1 lists the certified bounds in the 67 dimensions with a claimed improvement: 11–23, 25–40, 43–57, 62–75, 83–91, together with comparisons in dimensions 9 and 10. For $9 \le n \le 16$, the comparison values are the rigorous three-point bounds in Cohn–de Laat–Salmon [3, Table 1.1]. Tutunov et al. [7, Table 1] report slightly smaller numerical values; we use the rigorously verified values of [3] for comparison. For $n \ge 17$, we use numerical estimates from the highest available truncations of the dual solutions in [1] (in dimensions 17–91, the estimates from the two largest truncations differ by less than 10^{-20}).

The lower-bound column records the best lattice densities reported in the Nebe–Sloane catalogue [8], with the improvements in dimensions 38–40 from [9], and the entries in dimensions 54, 56, 64, and 72 from [10, 11]. We convert center density δ to packing density $v_n\delta$ and round down to six significant digits. Nonlattice records are excluded; a dash means that no sufficiently substantiated lattice-record value is supplied.

For $n = 9$, Dutour Sikirić and van Woerden's preprint [12, Corollary 1] establishes $\Delta_9^{\mathrm{lat}} = \sqrt{2}\,\pi^4/945$. Together with Mordell's inequality, this yields $\Delta_{10}^{\mathrm{lat}} \le \pi^5/(120 \cdot 2^{35/8}) < 0.122903$, so our bounds do not improve the strongest available lattice bounds in dimensions 9 and 10.

Table 1: Lattice packing density bounds. The comparison bounds are rigorous three-point bounds for $9 \le n \le 16$ and numerical Cohn–Elkies values for $n \ge 17$. Gains are relative to this comparison column.

n	Best known lattice	Our bound	Three-point bound/ Cohn–Elkies bound	Gain (%)
9	0.145774	0.184380634	0.191121	3.527
10	0.0920211	0.132492965	0.143411	7.613
11	0.0604326	0.0949892088	0.106726	11.00
12	0.0494541	0.0680145536	0.079712	14.67
13	0.0292084	0.0488231948	0.060165	18.85
14	0.0216240	0.0352392080	0.045062	21.80
15	0.0168575	0.0256118267	0.033757	24.13
16	0.0147081	0.0187461206	0.024995	25.00
17	0.00881131	0.0138839313	0.01846409034	24.81
18	0.00592836	0.0103704834	0.01348534045	23.10
19	0.00412080	0.00782447361	0.009817955140	20.30
20	0.00322586	0.00595808443	0.007127053603	16.40
21	0.00246588	0.00456045297	0.005159660395	11.61
22	0.00212766	0.00348596988	0.003725941969	6.441
23	0.00190532	0.00264134013	0.002684279886	1.600
25	0.000677212	0.00132930810	0.001384190722	3.965
26	0.000269220	0.000915119176	0.0009910238892	7.659
27	0.000128675	0.000631342154	0.0007082297959	10.86
28	6.97587×10^{-5}	0.000437000598	0.0005052542161	13.51
29	2.78789×10^{-5}	0.000303842544	0.0003598581852	15.57
30	1.44286×10^{-5}	0.000212402597	0.0002559028744	17.00
31	1.18377×10^{-5}	0.000149408588	0.0001817083814	17.78
32	1.10407×10^{-5}	0.000105764483	0.0001288432888	17.91
33	4.14068×10^{-6}	$7.55167890 \times 10^{-5}$	$9.123560390 \times 10^{-5}$	17.23
34	1.76697×10^{-6}	$5.43081491 \times 10^{-5}$	$6.452219674 \times 10^{-5}$	15.83
35	9.46190×10^{-7}	$3.93266737 \times 10^{-5}$	$4.557438431 \times 10^{-5}$	13.71

n	Best known lattice	Our bound	Three-point bound/ Cohn–Elkies bound	Gain (%)
36	6.16146×10^{-7}	$2.86680609 \times 10^{-5}$	$3.215305533 \times 10^{-5}$	10.84
37	3.21356×10^{-7}	$2.09981537 \times 10^{-5}$	$2.265869001 \times 10^{-5}$	7.328
38	2.17566×10^{-7}	$1.52662254 \times 10^{-5}$	$1.595064991 \times 10^{-5}$	4.291
39	1.68028×10^{-7}	$1.09486830 \times 10^{-5}$	$1.121686870 \times 10^{-5}$	2.391
40	1.47983×10^{-7}	$7.82816106 \times 10^{-6}$	$7.880105170 \times 10^{-6}$	0.6592
43	4.01571×10^{-8}	$2.70203239 \times 10^{-6}$	$2.716907473 \times 10^{-6}$	0.5475
44	3.20085×10^{-8}	$1.86230595 \times 10^{-6}$	$1.901770314 \times 10^{-6}$	2.075
45	2.52315×10^{-8}	$1.28034519 \times 10^{-6}$	$1.330090566 \times 10^{-6}$	3.740
46	2.27179×10^{-8}	$8.80676408 \times 10^{-7}$	$9.295151556 \times 10^{-7}$	5.254
47	2.14660×10^{-8}	$6.07126117 \times 10^{-7}$	$6.490757337 \times 10^{-7}$	6.463
48	2.31782×10^{-8}	$4.19948220 \times 10^{-7}$	$4.529067790 \times 10^{-7}$	7.277
49	—	$2.91698991 \times 10^{-7}$	$3.157963968 \times 10^{-7}$	7.631
50	—	$2.03458140 \times 10^{-7}$	$2.200392958 \times 10^{-7}$	7.536
51	—	$1.42487373 \times 10^{-7}$	$1.532136244 \times 10^{-7}$	7.001
52	—	$1.00209490 \times 10^{-7}$	$1.066123391 \times 10^{-7}$	6.006
53	—	$7.07065699 \times 10^{-8}$	$7.413767463 \times 10^{-8}$	4.628
54	1.46699×10^{-10}	$4.97631903 \times 10^{-8}$	$5.152285374 \times 10^{-8}$	3.415
55	—	$3.48109436 \times 10^{-8}$	$3.578475170 \times 10^{-8}$	2.721
56	5.35530×10^{-11}	$2.44093782 \times 10^{-8}$	$2.483937099 \times 10^{-8}$	1.731
57	—	$1.71634335 \times 10^{-8}$	$1.723196004 \times 10^{-8}$	0.3977
62	—	$2.74632309 \times 10^{-9}$	$2.746644716 \times 10^{-9}$	0.01171
63	—	$1.89257793 \times 10^{-9}$	$1.899489162 \times 10^{-9}$	0.3638
64	1.32606×10^{-12}	$1.29978772 \times 10^{-9}$	$1.312998022 \times 10^{-9}$	1.006
65	—	$8.92325642 \times 10^{-10}$	$9.071711718 \times 10^{-10}$	1.636
66	—	$6.13061198 \times 10^{-10}$	$6.264943126 \times 10^{-10}$	2.144
67	—	$4.21802530 \times 10^{-10}$	$4.324664005 \times 10^{-10}$	2.466
68	—	$2.90120314 \times 10^{-10}$	$2.984004045 \times 10^{-10}$	2.775
69	—	$2.00732130 \times 10^{-10}$	$2.058081299 \times 10^{-10}$	2.466
70	—	$1.38650284 \times 10^{-10}$	$1.418880624 \times 10^{-10}$	2.282
71	—	$9.60867646 \times 10^{-11}$	$9.778075467 \times 10^{-11}$	1.732
72	1.45861×10^{-13}	$6.66598519 \times 10^{-11}$	$6.735796328 \times 10^{-11}$	1.036
73	—	$4.62232194 \times 10^{-11}$	$4.638272400 \times 10^{-11}$	0.3439
74	—	$3.18709121 \times 10^{-11}$	$3.192705609 \times 10^{-11}$	0.1759
75	—	$2.19646063 \times 10^{-11}$	$2.196849358 \times 10^{-11}$	0.01769
83	—	$1.08926359 \times 10^{-12}$	$1.090200461 \times 10^{-12}$	0.08594
84	—	$7.47188544 \times 10^{-13}$	$7.478849669 \times 10^{-13}$	0.09312
85	—	$5.10524957 \times 10^{-13}$	$5.128979896 \times 10^{-13}$	0.4627
86	—	$3.48807350 \times 10^{-13}$	$3.516392699 \times 10^{-13}$	0.8053
87	—	$2.38928356 \times 10^{-13}$	$2.410106753 \times 10^{-13}$	0.8640
88	—	$1.63824494 \times 10^{-13}$	$1.651391388 \times 10^{-13}$	0.7961
89	—	$1.12348125 \times 10^{-13}$	$1.131203657 \times 10^{-13}$	0.6827
90	—	$7.71724614 \times 10^{-14}$	$7.746593987 \times 10^{-14}$	0.3788
91	—	$5.30260966 \times 10^{-14}$	$5.303494178 \times 10^{-14}$	0.01668

References

[1] N. Afkhami-Jeddi, H. Cohn, T. Hartman, D. de Laat, and A. Tajdini, *High-dimensional sphere packing and the modular bootstrap*, J. High Energy Phys. (2020), no. 12, 066; arXiv:2006.02560. Data: `https://hdl.handle.net/1721.1/125646`.

[2] H. Cohn and N. Elkies, *New upper bounds on sphere packings I*, Ann. of Math. **157** (2003), 689–714; arXiv:math/0110009.

[3] H. Cohn, D. de Laat, and A. Salmon, *Three-point bounds for sphere packing*, arXiv:2206.15373v1 (2022).

[4] H. Cohn and A. Kumar, *Optimality and uniqueness of the Leech lattice among lattices*, Ann. of Math. **170** (2009), 1003–1050; arXiv:math/0403263.

[5] J. Martinet, *Reduction modulo 2 and 3 of Euclidean lattices*, J. Algebra **251** (2002), 864–887. Author's version: `https://jamartin.perso.math.cnrs.fr/Publications/Latmod.pdf`.

[6] J. Martinet, *Lattices modulo 2 and 3 (comments)*, updated March 5, 2022, `https://jamartin.perso.math.cnrs.fr/Publications/latmodcomm.pdf`.

[7] R. Tutunov, A. Maraval, A. Grosnit, X. Li, J. Wang, and H. Bou-Ammar, *Model-Based and Sample-Efficient AI-Assisted Math Discovery in Sphere Packing*, arXiv:2512.04829v2 (2025).

[8] G. Nebe and N. J. A. Sloane, *Table of densest packings presently known*, Catalogue of Lattices, `https://www.math.rwth-aachen.de/~Gabriele.Nebe/LATTICES/density.html`.

[9] I. Dorofeev, X. Sun, and C. Wang, *Optimal extensions of cross-sections: Sphere packings in dimensions 38 to 43*, arXiv:2607.20359v4 (2026).

[10] G. Leterrier, *On the Mordell–Weil lattice of* $y^2 = x^3 + bx + t^{3^n+1}$ *in characteristic 3*, Res. Number Theory **8** (2022), article 23, doi:10.1007/s40993-022-00321-0.

[11] H. Cohn, *Table of sphere packing density bounds*, `https://cohn.mit.edu/sphere-packing/`.

[12] M. Dutour Sikirić and W. van Woerden, *The lattice packing problem in dimension 9 by Voronoi's algorithm*, preprint (2025), arXiv:2508.20719.